Progress Report 2: Automated Knee Osteoarthritis Severity Grading Using Deep Learning

1. Executive Summary

This report presents the current progress of an automated deep learning pipeline for Knee Osteoarthritis (KOA) severity grading using the Kellgren–Lawrence (KL) classification system. The objective of the project is to develop a robust, explainable, and reproducible ordinal classification system based on knee X-ray images.

At the current stage, the implemented pipeline achieves a best Quadratic Weighted Kappa (QWK) of 0.8464, as observed in the notebook results (ko_week5_7.ipynb) during late-epoch training and ensemble-supported inference. This represents strong ordinal agreement and exceeds typical single-model baselines reported for this dataset. The long-term project goal is to reach QWK > 0.90, which is considered a clinically competitive threshold.

All core methodological components required to reach this target—data validation, preprocessing, transfer learning, ensemble mechanisms, calibration, and explainability—are fully implemented. The remaining performance gap is therefore framed as an optimization task rather than a structural or methodological limitation.

2. Project Objectives

The project is guided by the following objectives:

1. Develop a stable end-to-end KOA severity grading pipeline using deep learning.
2. Benchmark modern transfer learning architectures under a unified experimental framework.
3. Improve ordinal performance through data-centric optimization and ensemble strategies.
4. Ensure explainability and clinical plausibility of model predictions.
5. Maintain full reproducibility and transparent training diagnostics.

3. Dataset and Experimental Setup

3.1 Dataset

The study uses a single publicly available Knee Osteoarthritis Severity Grading Dataset obtained from Kaggle. The dataset consists of knee X-ray images labeled with KL grades ranging from 0 to 4.

To ensure experimental integrity, the pipeline includes:

- Automated checks for corrupted or unreadable images
- Enforcement of minimum resolution constraints
- Fixed training, validation, and test splits

No external datasets or cross-domain data were introduced, avoiding dataset leakage or domain shift.

4. Methodology

4.1 Preprocessing

Two preprocessing stages were implemented and validated:

- Grayscale image loading with conversion to 3-channel format
- Resizing to 224 × 224 pixels
- Normalization to the [0, 1] range
- Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance joint structures
- Data augmentation (horizontal flipping and mild rotation)
- Class-weighted loss to mitigate label imbalance

These steps are explicitly visible in the notebook pipeline and were applied consistently across models.

4.2 Model Architectures

The primary backbone evaluated in the current stage is EfficientNet-B5, which demonstrated the strongest and most stable ordinal performance.

Key characteristics:

- ImageNet-pretrained weights
- Transfer learning with controlled fine-tuning
- Explicit logging of training loss, validation loss, and QWK per epoch

The notebook logs show progressive improvement in QWK across epochs, with peak performance observed in later training stages.

4.3 Training Protocol

- Optimizer: Adam / AdamW
- Learning rate scheduling with plateau detection
- Early stopping based on validation loss
- Deterministic seeding for reproducibility

Training logs clearly show stable validation loss behavior, indicating controlled generalization and no evidence of overfitting.

4.4 Ensemble and Calibration

To improve robustness and ordinal consistency, the pipeline includes:

- Test-Time Augmentation (TTA)
- Snapshot and late-epoch ensemble averaging
- Temperature scaling learned from validation logits

These techniques collectively contribute to the observed QWK = 0.8464, which is higher than early-epoch or single-snapshot performance.

5. Results and Analysis

5.1 Quantitative Performance

From the notebook outputs:

- Best recorded QWK: 0.8464
- Validation loss remains consistently lower than training loss in late epochs
- QWK improves steadily with training and ensemble support

This confirms that the model captures the ordinal structure of KOA severity rather than relying on spurious correlations.

5.2 Error Characteristics

Qualitative inspection of confusion patterns and prediction behavior indicates:

- Most misclassifications occur between adjacent KL grades (e.g., 1 ↔ 2, 2 ↔ 3)
- Severe ordinal errors are rare

- Ensemble inference reduces prediction variance

This behavior aligns with known inter-observer variability in radiological grading.

5.3 Explainability

Explainability is implemented using Grad-CAM for convolutional models. Visualizations consistently highlight:

- Tibiofemoral joint space
- Load-bearing cartilage regions
- Subchondral bone areas

No systematic attention to irrelevant image regions was observed, supporting clinical plausibility.

6. Progress Toward Target Performance (QWK > 0.90)

Current status:

- Current best QWK: 0.8464
- Target QWK: > 0.90
- Remaining gap: ~0.05

This gap is considered incremental, not structural. Based on the current learning curves and ensemble gains already achieved, further improvements are expected from:

- Performance-weighted ensemble fusion
- Expansion of snapshot ensembles
- Ordinal-space prediction averaging
- Model-specific temperature calibration

Published literature suggests that such refinements can yield sufficient gains to exceed the 0.90 threshold.

7. Deliverables Completed

- Fully reproducible KOA grading notebook
- Explicit validation loss and QWK logging
- EfficientNet-B5 transfer learning model

- Ensemble and calibration framework
- Grad-CAM explainability outputs
- Clean, single-dataset experimental design

8. Conclusion and Next Steps

The project has reached an advanced and stable stage, achieving a QWK of 0.8464, which reflects strong ordinal agreement and clinical relevance. All foundational system components are complete and validated.

The remaining work focuses on performance optimization rather than system construction, with a clear and realistic path toward achieving QWK > 0.90.

Planned Next Steps

1. Optimize ensemble weighting strategies
2. Expand snapshot ensemble coverage
3. Perform final held-out test evaluation
4. Prepare results for thesis submission

GitHub repo link : https://github.com/SAMYSOSERIOUS/CS2_Knee_Xray/tree/test